1. **Which one of the following statements is true regarding K Nearest Neighbors?** 1 point

- ◯ The Manhattan distance between two data points is the square root of the sum of the squares of the differences between the individual feature values of the data points.
- ◯ For high dimensional data, the best distance measure to use for KNN is the Euclidean distance.
- ◯ K Nearest Neighbors (KNN) assumes that points which are close together are similar.
- ◯ The distance between two data points is independent of the scale of their features.

2. **Which one of the following statements is most accurate?** 1 point

- ◯ KNN only needs to remember the hyperplane coefficients to classify a new data sample.
- ◯ KNN determines which points are closest to a given data point, so it doesn't take long to actually perform prediction.
- ◯ K nearest neighbors (KNN) needs to remember the entire training dataset in order to classify a new data sample.
- ◯ Linear regression needs to remember the entire training dataset in order to make a prediction for a new data sample.

3. **Which one of the following statements is most accurate about K Nearest Neighbors (KNN)?** 1 point

- ◯ KNN is a regression model.
- ◯ KNN is a classification model.
- ◯ KNN can be used for both classification and regression.
- ◯ KNN is an unsupervised learning method.

4. **(True/False) K Nearest Neighbors with large *k* tend to be the best classifiers.** 1 point

- ◯ True
- ◯ False